

```
import pandas as pd
import numpy as np
```

```
# What are vectorized operations
a = np.array([1,2,3,4])
a * 4
```

```
array([ 4,  8, 12, 16])
```

```
# problem in vectorized operations in vanilla python
s = ['cat', 'mat', None, 'rat']
```

```
[i.startswith('c') for i in s]
```

```
-----
AttributeError                                Traceback (most recent call last)
<ipython-input-19-3fe713c7ebb8> in <module>
      2 s = ['cat', 'mat', None, 'rat']
      3
----> 4 [i.startswith('c') for i in s]

<ipython-input-19-3fe713c7ebb8> in <listcomp>(.0)
      2 s = ['cat', 'mat', None, 'rat']
      3
----> 4 [i.startswith('c') for i in s]

AttributeError: 'NoneType' object has no attribute 'startswith'
```

```
# How pandas solves this issue?
```

```
s = pd.Series(['cat', 'mat', None, 'rat'])
# string accessor
s.str.startswith('c')
```

```
# fast and optimized
```

```
0    True
1    False
2     None
3    False
dtype: object
```

```
# import titanic
df = pd.read_csv('/content/titanic.csv')
df['Name']
```

```
0      Braund, Mr. Owen Harris
1  Cumings, Mrs. John Bradley (Florence Briggs Th...
2      Heikkinen, Miss. Laina
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)
4      Allen, Mr. William Henry
...
886      Montvila, Rev. Juozas
887      Graham, Miss. Margaret Edith
888  Johnston, Miss. Catherine Helen "Carrie"
889      Behr, Mr. Karl Howell
890      Dooley, Mr. Patrick
Name: Name, Length: 891, dtype: object
```

```
# Common Functions
```

```
# lower/upper/capitalize/title
```

```
df['Name'].str.upper()
```

```
df['Name'].str.capitalize()
```

```
df['Name'].str.title()
```

```
# len
```

```
df['Name'][df['Name'].str.len() == 82].values[0]
```

```
# strip
```

```
"          nitish                ".strip()
```

```
df['Name'].str.strip()
```

```
0      Braund, Mr. Owen Harris
1  Cumings, Mrs. John Bradley (Florence Briggs Th...
2      Heikkinen, Miss. Laina
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)
```

```

4             Allen, Mr. William Henry
...
886             Montvila, Rev. Juozas
887             Graham, Miss. Margaret Edith
888             Johnston, Miss. Catherine Helen "Carrie"
889             Behr, Mr. Karl Howell
890             Dooley, Mr. Patrick
Name: Name, Length: 891, dtype: object

```

```

# split -> get
df['lastname'] = df['Name'].str.split(',').str.get(0)
df.head()

```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	lastname
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	Braund
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C	Cumings
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	Heikkinen
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lilv Mav Peel)	female	35.0	1	0	113803	53.1000	C123	S	Futrelle

```

df[['title','firstname']] = df['Name'].str.split(',').str.get(1).str.strip().str.split(' ', n=1, expand=True)
df.head()

```

```
df['title'].value_counts()
```

```

Mr.      517
Miss.    182
Mrs.     125
Master.   40
Dr.        7
Rev.        6
Mlle.        2
Major.        2
Col.        2
the          1
Capt.        1
Ms.          1
Sir.          1
Lady.         1
Mme.          1
Don.          1
Jonkheer.     1
Name: title, dtype: int64

```

```

# replace
df['title'] = df['title'].str.replace('Ms.','Miss.')
df['title'] = df['title'].str.replace('Mlle.','Miss.')

```

```

<ipython-input-60-9403e5934d1f>:2: FutureWarning: The default value of regex will change from True to False in a future version.
df['title'] = df['title'].str.replace('Ms.','Miss.')
<ipython-input-60-9403e5934d1f>:3: FutureWarning: The default value of regex will change from True to False in a future version.
df['title'] = df['title'].str.replace('Mlle.','Miss.')

```

```
df['title'].value_counts()
```

```

Mr.      517
Miss.    185
Mrs.     125
Master.   40
Dr.        7
Rev.        6
Major.        2
Col.        2
Don.         1
Mme.         1
Lady.         1
Sir.         1
Capt.        1
the          1
Jonkheer.     1
Name: title, dtype: int64

```

```

# filtering
# startswith/endswith

```

```
df[df['firstname'].str.endswith('A')]
# isdigit/isalpha...
df[df['firstname'].str.isdigit()]
```



PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	lastname	title	firstname
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------	----------	-------	-----------

```
# applying regex
# contains
# search john -> both case
df[df['firstname'].str.contains('john',case=False)]
# find lastnames with start and end char vowel
df[df['lastname'].str.contains('^[^aeiouAEIOU].+[^aeiouAEIOU]$')]
```



PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	lastname	title	firstname	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	Braund	Mr.	Owen Harris
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C	Cumings	Mrs.	John Bradley (Florence Briggs Thayer)
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	Heikkinen	Miss.	Laina
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	NaN	Q	Moran	Mr.	James
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S	McCarthy	Mr.	Timothy J
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
884	885	0	3	Sutehall, Mr. Henry Jr	male	25.0	0	0	SOTON/OQ 392076	7.0500	NaN	S	Sutehall	Mr.	Henry Jr

```
# slicing
df['Name'].str[::-1]

0      sirraH newO .rM ,dnuaR
1    )reyaHT sggirB ecnerolF( yeldarB nhoJ .srM ,sg...
2      aniaL .ssiM ,nenikkieH
3    )leeP yaM yliL( htaeH seuqcaJ .srM ,ellertuF
4      yrneH mailliW .rM ,nella
...
886      sazouJ .veR ,alivtnoM
887      htidE teragraM .ssiM ,maharG
888    "eirraC" neleH enirehtaC .ssiM ,notsnhoJ
889      llewoH lraK .rM ,rheB
890      kcirtaP .rM ,yelooD
Name: Name, Length: 891, dtype: object
```

Start coding or [generate](#) with AI.

